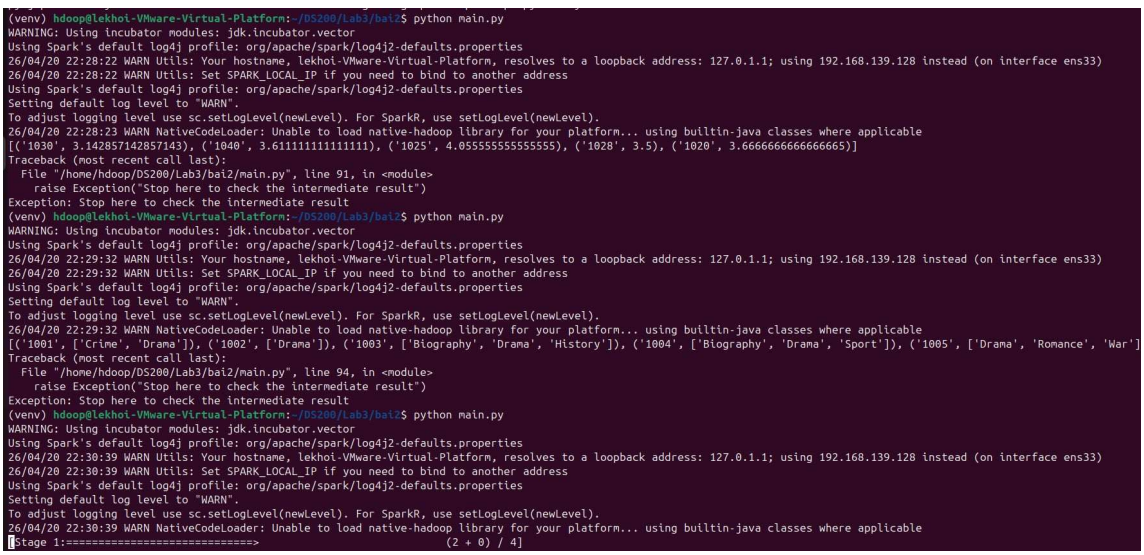
Tên : Lê Minh Khôi – 23520767

Lab3 – DS200.Q21

Báo cáo hình kết quả Lab 3

```
Movie with the highest average rating:
[('1015', ('Sunset Boulevard (1950)', (7, 4.357142857142857)))]
(venv) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab3/bai1$
```

Figure 1. Kết quả bài 1



```
(venv) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab3/bai2$ python main.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/04/20 22:28:22 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/04/20 22:28:22 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/20 22:28:23 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
[('1030', 3.142857142857143), ('1040', 3.611111111111111), ('1025', 4.055555555555555), ('1028', 3.5), ('1020', 3.6666666666666665)]
Traceback (most recent call last):
  File "/home/hdoop/DS200/Lab3/bai2/main.py", line 91, in <module>
    raise Exception("Stop here to check the intermediate result")
Exception: Stop here to check the intermediate result
(venv) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab3/bai2$ python main.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/04/20 22:29:32 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/04/20 22:29:32 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/20 22:29:32 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
[('1001', ['Crime', 'Drama']), ('1002', ['Drama']), ('1003', ['Biography', 'Drama', 'History']), ('1004', ['Biography', 'Drama', 'Sport']), ('1005', ['Drama', 'Romance', 'War']
Traceback (most recent call last):
  File "/home/hdoop/DS200/Lab3/bai2/main.py", line 94, in <module>
    raise Exception("Stop here to check the intermediate result")
Exception: Stop here to check the intermediate result
(venv) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab3/bai2$ python main.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/04/20 22:30:39 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/04/20 22:30:39 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/20 22:30:39 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
[Stage 1:============================>                            (2 + 0) / 4]
```

Figure 2. Kết quả bài 2

- Bài 2 em không biết tại sao đến bước join thì máy chạy mãi không xong. Em nghĩ do máy ảo của em quá yếu bởi vì em kiểm tra output của các bước trước thì hoàn toàn bình thường, cấu trúc 2 RDD để join cũng hợp lý nhưng đến bước join thì chạy mãi không xong

```
Step 1 - users_gender_rdd:
[('1', 'M'), ('2', 'F'), ('3', 'M'), ('4', 'F'), ('5', 'M')]

Step 2 - ratings_parsed_rdd:
[('7', ('1020', 4.5)), ('23', ('1015', 3.5)), ('45', ('1030', 4.0)), ('12', ('1047', 3.0)), ('38', ('1012', 4.5))]

Step 3 - joined_user_rating_rdd:
[('4', ('F', ('1037', 4.0))), ('5', ('M', ('1013', 4.5))), ('5', ('M', ('1015', 4.5))), ('5', ('M', ('1037', 4.0))), ('5', ('M', ('1050', 4.5)))]

Step 4 - movie_gender_rating_rdd:
[(('1037', 'F'), (4.0, 1)), (('1013', 'M'), (4.5, 1)), (('1015', 'M'), (4.5, 1)), (('1037', 'M'), (4.0, 1)), (('1050', 'M'), (4.5, 1))]

Step 5 - movie_gender_sum_count_rdd:
[(('1025', 'F'), (45.5, 11)), (('1028', 'F'), (10.5, 3)), (('1040', 'F'), (25.5, 7)), (('1040', 'M'), (39.5, 11)), (('1025', 'M'), (27.5, 7))]

Step 6 - movie_gender_avg_rdd:
(('1025', 'F'), 4.136363636363637)
(('1028', 'F'), 3.5)
(('1040', 'F'), 3.642857142857143)
(('1040', 'M'), 3.590909090909091)
(('1025', 'M'), 3.9285714285714284)
(('1028', 'M'), 3.5)
(('1037', 'F'), 3.8)
(('1013', 'M'), 4.055555555555555)
(('1015', 'M'), 4.333333333333333)
(('1037', 'M'), 4.0)
(('1050', 'M'), 4.0)
(('1039', 'M'), 3.75)
(('1020', 'F'), 3.55)
(('1013', 'F'), 3.9375)
(('1050', 'F'), 3.3214285714285716)
(('1020', 'M'), 3.8125)
(('1012', 'F'), 4.0)
(('1039', 'F'), 3.9)
(('1047', 'F'), 3.6666666666666665)
(('1015', 'F'), 4.5)
(venv) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab3/bai3$
```

Figure 3. Kết quả bài 3

```
Step 1 - users_age_group_rdd:
[('1', '26-35'), ('2', '26-35'), ('3', '36-45'), ('4', '18-25'), ('5', '26-35')]

Step 2 - ratings_parsed_rdd:
[('7', ('1020', 4.5)), ('23', ('1015', 3.5)), ('45', ('1030', 4.0)), ('12', ('1047', 3.0)), ('38', ('1012', 4.5))]

Step 3 - joined_rdd:
[('4', ('18-25', ('1037', 4.0))), ('5', ('26-35', ('1013', 4.5))), ('5', ('26-35', ('1015', 4.5))), ('5', ('26-35', ('1037', 4.0))), ('5', ('26-35', ('

Step 4 - movie_agegroup_rating_rdd:
[(('1037', '18-25'), (4.0, 1)), (('1013', '26-35'), (4.5, 1)), (('1015', '26-35'), (4.5, 1)), (('1037', '26-35'), (4.0, 1)), (('1050', '26-35'), (4.5,

Step 5 - movie_agegroup_sum_count_rdd:
[(('1020', '36-45'), (30.5, 8)), (('1037', '36-45'), (33.5, 9)), (('1013', '36-45'), (34.0, 8)), (('1039', '36-45'), (27.0, 7)), (('1025', '18-25'), (4

Step 6 - movie_agegroup_avg_rdd:
(('1020', '36-45'), 3.8125)
(('1037', '36-45'), 3.7222222222222223)
(('1013', '36-45'), 4.25)
(('1039', '36-45'), 3.857142857142857)
(('1025', '18-25'), 4.5)
(('1010', '26-35'), 3.6)
(('1037', '46-55'), 4.166666666666667)
(('1043', '26-35'), 3.857142857142857)
(('1015', '46-55'), 4.5)
(('1039', '46-55'), 3.5)
(('1047', '36-45'), 3.5)
(('1012', '36-45'), 3.5)
(('1050', '36-45'), 3.642857142857143)
(('1015', '36-45'), 4.5)
(('1040', '18-25'), 3.5)
(('1047', '46-55'), 4.0)
(('1020', '46-55'), 4.0)
(('1037', '18-25'), 4.0)
(('1040', '36-45'), 3.857142857142857)
(('1030', '26-35'), 3.0)
(venv) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab3/bai4$
```

Figure 4. Kết quả bài 4

```
(venv) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab3/bai5$ python main.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/04/21 07:24:29 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens3
26/04/21 07:24:29 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/21 07:24:30 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Step 1 - users_occ_rdd:
[('1', '3'), ('2', '7'), ('3', '2'), ('4', '10'), ('5', '1')]

Step 2 - ratings_parsed_rdd:
[('7', 4.5), ('23', 3.5), ('45', 4.0), ('12', 3.0), ('38', 4.5)]

Step 3 - joined_rdd:
[('4', ('10', 4.0)), ('5', ('1', 4.5)), ('5', ('1', 4.5)), ('5', ('1', 4.0)), ('5', ('1', 4.5))]

Step 4 - occ_rating_rdd:
[('10', (4.0, 1)), ('1', (4.5, 1)), ('1', (4.5, 1)), ('1', (4.0, 1)), ('1', (4.5, 1))]

Step 5 - occ_sum_count_rdd:
[('10', (21.5, 6)), ('5', (62.0, 17)), ('4', (18.5, 5)), ('1', (42.5, 10)), ('2', (77.5, 21))]

Step 6 - occupation_map_rdd:
[('1', 'Programmer'), ('2', 'Doctor'), ('3', 'Engineer'), ('4', 'Teacher'), ('5', 'Lawyer')]

Step 7 - joined_occ_rdd:
[Stage 15:=====================================>                   (4 + 0) / 6]
```

Figure 5. Kết quả bài 5

- Bài 5 cũng gặp vấn đề tương tự như bài 2 khi có lệnh join giữa 2 RDD lớn

```
(venv) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab3/bai6$ python main.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/04/21 07:33:03 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens3
26/04/21 07:33:03 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/21 07:33:04 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/04/21 07:33:05 WARN Utils: Service 'SparkUI' could not bind on port 4040. Attempting port 4041.
Step 1 - ratings_rdd:
['7,1020,4.5,1577836800', '23,1015,3.5,1577923200', '45,1030,4.0,1578009600', '12,1047,3.0,1578096000', '38,1012,4.5,1578182400']

Step 2 - year_rating_rdd:
[(2020, (4.5, 1)), (2020, (3.5, 1)), (2020, (4.0, 1)), (2020, (3.0, 1)), (2020, (4.5, 1))]

Step 3 - year_sum_count_rdd:
[(2020, (690.5, 184))]

Final result - (Year, total_count, avg_rating):
(2020, 184, 3.7527173913043477)
(venv) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab3/bai6$
```

Figure 6. Kết quả bài 6